

MENGZI, China

WMO: 569850

Lat: 23.4444N

Lon: 103.3278E

Elev: 4310

StdP: 12.55

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.3	43.0	29.2	27.2	63.4	32.8	32.0	59.8	20.7	63.1	18.2	63.1	3.7	180	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.4	87.8	65.0	86.0	66.0	84.4	66.5	71.6	80.6	70.9	79.6	70.3	78.7	8.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	126.6	74.2	68.6	123.6	74.0	68.1	121.2	73.8	38.5	80.6	37.9	79.9	37.4	78.9	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	14.4	12.3	DB	35.9	91.1	3.6	1.9	33.3	92.5	31.2	93.6	29.2	94.6	26.6	96.0	(4)
			WB	34.0	72.6	3.5	1.1	31.5	73.4	29.4	74.0	27.5	74.6	24.9	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.1	56.5	60.7	66.7	72.0	74.3	75.7	75.0	74.5	72.7	68.7	62.8	57.0
	DBStd	8.47	5.95	7.46	6.52	5.54	5.03	3.39	2.70	2.77	3.55	4.25	4.80	5.26
	HDD50	44	18	13	2	0	0	0	0	0	0	0	1	10
	HDD65	856	266	148	60	12	5	0	0	0	3	19	92	251
	CDD50	6645	219	312	521	661	752	771	775	760	682	580	386	226
	CDD65	1984	2	28	114	223	292	322	310	295	235	134	27	2
	CDH74	10288	17	169	712	1497	1894	1816	1420	1317	958	395	85	8
	CDH80	2129	1	11	130	437	548	382	242	224	130	22	2	0
Wind	WSAvg	5.7	6.3	7.0	7.1	7.2	6.4	5.5	4.7	4.0	4.5	5.0	5.0	5.4
Precipitation	PrecAvg	33.2	0.6	0.6	0.9	2.0	3.5	4.9	6.5	6.1	3.5	2.2	1.6	0.7
	PrecMax	47.0	2.6	2.8	3.5	5.6	8.2	13.2	12.6	12.7	7.7	9.4	6.4	4.7
	PrecMin	20.1	0.0	0.0	0.0	0.4	0.9	1.4	1.7	0.7	0.9	0.0	0.0	0.0
	PrecStd	6.1	0.6	0.7	0.9	1.2	1.9	2.3	2.5	2.4	1.5	1.8	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	81.7	86.3	89.5	90.7	88.1	87.1	86.8	85.4	82.4	79.4	74.8
		MCWB	53.7	55.6	57.4	61.1	64.4	69.2	69.8	69.8	68.8	66.7	61.5	57.3
	2%	DB	72.9	78.3	83.5	87.3	88.0	85.9	84.7	84.5	83.3	80.1	76.5	71.9
		MCWB	54.4	55.3	57.7	61.7	65.2	69.2	69.6	69.9	68.4	65.7	60.8	56.9
	5%	DB	70.2	76.0	81.1	85.0	85.6	84.1	82.8	82.5	81.4	78.2	73.9	69.1
		MCWB	54.2	55.5	57.8	62.2	65.9	69.2	69.7	69.5	67.8	64.9	60.1	56.5
	10%	DB	67.2	73.2	78.3	82.4	83.1	82.2	80.8	80.6	79.5	76.2	71.5	66.2
		MCWB	54.0	55.3	58.1	62.4	66.3	69.0	69.5	69.2	67.3	64.4	59.3	55.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	61.1	64.0	68.2	70.5	72.0	72.4	72.6	71.2	68.7	65.3	61.2
		MCDB	67.1	72.5	75.0	79.5	82.2	82.1	81.6	81.7	79.9	78.1	73.9	70.4
	2%	WB	57.3	59.5	62.7	66.7	69.4	71.0	71.5	71.5	70.2	67.5	63.6	59.2
		MCDB	66.6	69.9	73.3	77.8	80.5	80.9	80.0	80.0	78.8	76.3	71.6	67.7
	5%	WB	55.9	58.3	61.6	65.8	68.8	70.4	70.8	70.8	69.4	66.6	62.3	57.5
		MCDB	65.8	68.7	72.2	76.7	79.2	79.8	78.8	78.7	77.6	74.7	69.9	66.3
	10%	WB	54.9	57.2	60.6	64.8	68.1	69.9	70.2	70.0	68.6	65.7	61.0	56.2
		MCDB	64.6	68.2	71.3	75.9	77.9	78.8	77.7	77.6	76.3	73.2	68.4	64.6
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.5	17.0	16.0	13.7	11.4	10.9	11.9	12.5	12.1	14.6	14.5
		MCDBR	21.3	21.4	21.2	19.1	16.7	14.0	14.1	15.0	15.1	15.4	18.1	19.4
		MCWBR	7.6	6.6	6.1	5.1	4.1	3.7	4.1	4.4	4.6	4.9	6.3	7.5
	5% WB	MCDBR	17.6	17.2	16.8	16.1	14.0	12.0	12.0	12.9	12.8	12.8	14.3	16.3
		MCWBR	7.3	6.2	6.2	5.3	4.4	3.8	4.3	4.6	4.4	4.5	5.8	7.0
Clear-Sky Solar Irradiance	taub	0.406	0.437	0.627	0.590	0.447	0.426	0.430	0.472	0.468	0.482	0.422	0.434	
	taud	2.167	2.089	1.620	1.770	2.253	2.346	2.316	2.173	2.160	2.062	2.171	2.062	
	Ebn at Noon	268	267	223	236	271	274	273	262	262	251	260	252	
	Edh at Noon	43	49	83	72	44	40	42	48	48	50	42	46	
All-Sky Solar Radiation	RadAvg	1096	1434	1558	1741	1665	1414	1379	1375	1308	1079	1164	985	
	RadStd	141	215	133	160	187	126	86	71	79	121	96	114	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.92	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	+347	+265
Regional (6 neighbors)		+0.67	N/A	N/A	+1.22	N/A	N/A	N/A	N/A	+191	+108

Nomenclature: See separate page